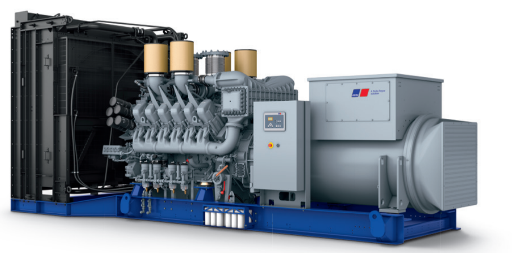
Diesel Generator Set

mtu 12V4000 DS2000

380V – 11 kV/50 Hz/standby power/NEA (ORDE) optimized 12V4000G84F/water charge air cooling



Optional equipment and finishing shown. Standard may vary.

Product highlights

Benefits
- Low fuel consumption
- Optimized system integration ability
- High reliability
- High availability of power
- Long maintenance intervals

Support
- Global product support offered

Standards
- Engine-generator set is designed and manufactured in facilities certified to standards ISO 2008:9001 and ISO 2004:14001
- Generator set complies to ISO 8528
- Generator meets NEMA MG1, BS 5000, ISO, DIN EN and IEC standards
- NFPA 110

Power rating
- System ratings: 1970 kVA - 2080 kVA
- Accepts rated load in one step per NFPA 110*
- Generator set complies to G3 according to ISO 8528-5
- Generator set exceeds load steps according to ISO 8528-5*

Performance assurance certification (PAC)
- Engine-generator set tested to ISO 8528-5 for transient response
- 85% load factor
- Verified product design, quality and performance integrity
- All engine systems are prototype and factory tested

Complete range of accessories available
- Control panel
- Power panel
- Circuit breaker/power distribution
- Fuel system
- Fuel connections with shut-off valve mounted to base frame
- Starting/charging system
- Exhaust system
- Mechanical and electrical driven radiators
- Medium and oversized voltage alternators

Emissions
- NEA (ORDE) optimized

Certifications
- CE certification option
- Unit certificate acc. to VDE-AR-N 4110

* Changes to the standard parameter sets (alternator-regulator and genset-controller) are necessary

A Rolls-Royce solution

Application data [1]

Engine

Manufacturer	*mtu*
Model	12V4000G84F
Type	4-cycle
Arrangement	12V
Displacement: l	57.2
Bore: mm	170
Stroke: mm	210
Compression ratio	16.4
Rated speed: rpm	1500
Engine governor	ECU 9
Max power: kWm	1750
Air cleaner	dry

Fuel system

Maximum fuel lift: m	5
Total fuel flow: l/min	16

Fuel consumption [2]	l/hr	g/kwh
At 100% of power rating:	419.6	199
At 75% of power rating:	316.9	200
At 50% of power rating:	220.3	209

Liquid capacity (lubrication)

Total oil system capacity: l	260
Engine jacket water capacity: l	160
Intercooler coolant capacity: l	40

Combustion air requirements

Combustion air volume: m^3/s	2.1
Max. air intake restriction: mbar	50

Cooling/radiator system

Coolant flow rate (HT circuit): m^3/hr	56.0
Coolant flow rate (LT circuit): m^3/hr	30
Heat rejection to coolant: kW	650
Heat radiated to charge air cooling: kW	360
Heat radiated to ambient: kW	75
Fan power for electr. radiator (40°C): kW	55

Exhaust system

Exhaust gas temp. (after turbocharger): °C	520
Exhaust gas volume: m^3/s	5.8
Maximum allowable back pressure: mbar	85
Minimum allowable back pressure: mbar	30

Standard and optional features

System ratings (kW/kVA)

Generator model	Voltage	NEA (ORDE) optimized					
		without radiator			with mechanical radiator		
		kWel	kVA*	AMPS	kWel	kVA*	AMPS
Leroy Somer LSA52.3 S6 (Low voltage Leroy Somer standard)	380 V	1664	2080	3160	1608	2010	3054
	400 V	1664	2080	3002	1608	2010	2901
	415 V	1664	2080	2894	1608	2010	2796
Leroy Somer LSA52.3 S7 (Low voltage Leroy Somer oversized)	380 V	1664	2080	3160	1608	2010	3054
	400 V	1664	2080	3002	1608	2010	2901
	415 V	1664	2080	2894	1608	2010	2796
Marathon 744RSL7091 (Low voltage Marathon)	380 V	1584	1980	3008	1576	1970	2993
	400 V	1624	2030	2930	1600	2000	2887
	415 V	1608	2010	2796	1600	2000	2782
Marathon 744RSL7092 (Low voltage Marathon oversized)	380 V	1584	1980	3008	1576	1970	2993
	400 V	1624	2030	2930	1600	2000	2887
	415 V	1608	2010	2796	1600	2000	2782
Marathon 1020FDH7096 (Medium volt. marathon)	11 kV	1656	2070	109	1600	2000	105
Leroy Somer LSA53.2 VL7 (Medium volt. Leroy Somer)	11 kV	1664	2080	109	1608	2010	105

* cos phi = 0.8

1 All data refers only to the engine and is based on ISO standard conditions (25°C and 100m above sea level).

2 Values referenced are in accordance with ISO 3046-1. Conversion calculated with fuel density of 0.83 g/ml. All fuel consumption values refer to rated engine power.